$\left(\frac{2a}{q}\right) = \left(\frac{2}{q}\right)\left(\frac{a}{q}\right)$; $\left(\frac{2}{q}\right) = (-1)^{\frac{q^2-1}{8}}$

$\left(\frac{1}{q}\right) = 1$

$q \in P'_{\mathbb{N}}$ $\left(\frac{a}{q}\right) = 1 \Leftrightarrow$ a - квадрат

$\left(\frac{a}{q}\right) = -1 \Leftrightarrow$ a не квадрат

$\left(\frac{a}{q}\right) = \left(\frac{a-q}{q}\right)$, $a \geq q$, $a \notin (2\mathbb{N})$

$\left(\frac{a}{q}\right) = (-1)^{\frac{a-1}{2}\cdot\frac{q-1}{2}} \cdot \left(\frac{q}{a}\right)$, $a < q$, $a \notin (2\mathbb{N})$

$q \notin P'_{\mathbb{N}}$ $\left(\frac{a}{q}\right) = 1 \Rightarrow$ a быть может не

$\left(\frac{a}{q}\right) = -1 \Rightarrow$ a - точно

Вычислить. $\left(\frac{1381}{1533}\right)$?

$= (-1)^{\frac{1380}{2}\cdot\frac{1382}{2}} \cdot \left(\frac{1533}{1381}\right) = -\left(\frac{152}{1381}\right) = \left(\frac{2^3}{1381}\right)\left(\frac{19}{1381}\right) = \left(\frac{19}{1381}\right) =$ ($(-1)^{\frac{1381^2-1}{8}} = -1$)

$= (-1)^{\frac{18}{2}\cdot\frac{5}{2}} \cdot \left(\frac{1381}{19}\right) = \left(\frac{1381}{19}\right) =$ ($(-1)^{345}$, $\frac{1380\cdot 1382}{8} = 691$)

$= \left(\frac{13}{19}\right) = (-1)^{\frac{12}{2}\cdot\frac{18}{2}} \cdot \left(\frac{19}{13}\right) = \left(\frac{6}{13}\right) = \left(\frac{2}{13}\right)\cdot\left(\frac{3}{13}\right) = \left(\frac{3}{13}\right) =$ ($(-1)^{\frac{132-1}{8}} = -1$, $\frac{12\cdot 14}{8} = 21$)

$= (-1)^{\frac{2}{2}\cdot\frac{4^2}{2}} \cdot \left(\frac{13}{3}\right) = \left(\frac{1}{3}\right) = 1$?

$1533 = 3\cdot 511 = 3\cdot 7\cdot 73 =$ (1) Ответ: быть

$\left(\frac{1381}{3}\right) = \left(\frac{1}{3}\right) = 1$

$\left(\frac{1381}{7}\right) = \left(\frac{2}{7}\right) = (-1)^{\frac{7^2-1}{8}} = (-1)^{\frac{3\cdot 8^2}{8}} = -1$

$\left(\frac{1381}{73}\right) = \left(\frac{67}{73}\right) = (-1)^{\frac{33}{2}\cdot\frac{34}{2}} \left(\frac{6}{67}\right) = \left(\frac{2}{67}\right)\left(\frac{3}{67}\right) =$

$= -\left(\frac{3}{67}\right) = (-1)^{\frac{2}{2}\cdot\frac{66^2}{8}} \cdot \left(\frac{67}{3}\right) = -1$ ($(-1)^{\frac{67^2-1}{8}} = -1$, $(-1)^{\frac{33\cdot 66\cdot 68}{8\cdot 2}}$)

1381 | 3
12 | 460
18
1

1381 | 7
7 | 197
68
63
51
49
2

67 | 3
6 | 22
7

1381 | 73
73·10 | 18
651
584
67